Number Theory

Lecture 2

22 January, 2025

1 Divisibility:

If a, b are integers with $b \neq 0$ and if there exists an integer x such that $a = bx$, then we say "a is divisible by b" and denoted by $b|a$.

1.1 Basic Properties

1. $b|a \Rightarrow -b|a$
2. $a|0; \quad 1|a; \quad a|a$
3. $a|b,\ b|c \Rightarrow a|c$
4. $a|b,\ c|d \Rightarrow ac|bd$
5. $a|b,\ a|c \Rightarrow a|bx + cy$ for any integer x, y
6. $a|b$ and $b \neq 0 \Rightarrow |a| \leq |b|$

Exercise. Prove that $8|\ 5^{2n} + 7$.

Solution. For $n = 1$, $\ 8|\ 25 + 7$ i.e. $8|32$, which is true.
Assume that, it's true for $n = k$, i.e. $8|\ 5^{2k} + 7$,so $5^{2k} + 7 = 8u$, for some u
Now,

$$\begin{aligned}
5^{2(k+1)} + 7 &= 5^{2k} \cdot 5^2 + 7 \\
&= 25(8u - 7) + 7 \\
&= 200u - 175 + 7 \\
&= 200u - 168 \\
&= 8(25u - 21)
\end{aligned}$$

i.e. $8|\ 5^{2(k+1)} + 7$
So, by mathematical induction, it is true for all integers.

1.2 Definition (Greatest Common Divisor)

Suppose, a and b are integers not both simultaneously zeros. A positive integer 'd' is called the greatest common divisor (gcd) of a and b, denoted by gcd(a,b) or (a,b) if

(i) $d|a$ and $d|b$
(ii) For any other integer with $c|a,\ c|d$, we have $c \leq d$.

1.3 Theorem: If $d =$ gcd(a,b), then there exists integers x and y such that $d = ax + by$.

Proof.
Let, $S = \{au + bv \mid u, v$ are integers such that $au + bv > 0\}$

$S \neq \phi$, as $|a| = a.(\text{sgn } a) + b.0 \in S$ if $a \neq 0$, $\quad |b| = a.0 + b.(\text{sgn } b) \in S$ if $b \neq 0$

By W.O.P, S has a least element d'(say).
Then, $d' > 0$ and $d' = ax + by$ for some integers x and y.

By division algorithm, there exists integers q and r such that $a = qd' + r, \quad 0 \leq r < d'$.
If $r \neq 0$, then

$$\begin{aligned} r &= a - qd' \\ &= a - q(ax + by) \\ &= (1 - qx)a + (-qy)b \\ &= au + bv \in S \end{aligned}$$

a contradiction to the fact that d' is least in S.
Hence, $r = 0$, so $d'|a$
Similarly, $d'|b$.
Also, if $c|a$ and $c|b$, then $c|ax + by$ so $c|d'$
So, $c \leq d'$
Hence, $d' = d =$ gcd(a,b)
i.e. gcd(a,b)=$ax + by$ for some integers x and y.

1.4 Definition

Suppose a and b are integers with gcd(a,b)=1, then a and b are said to be relatively prime.

1.5 Theorem: a, b are relatively prime $\iff$ there exist integers x and y such that $ax + by = 1$.

Proof.
Suppose, a, b are relatively prime, so gcd(a,b)=1.
Hence, by previous theorem, there exits integers x and y such that $ax + by = 1$.

Conversely, suppose, there exists integers x and y such that $ax + by = 1$.
Suppose, $d =$gcd(a,b)
Then, $d|a$ and $d|b$, hence $d|\ ax + by$, thus $d|1$ and so $d = 1$. Therefore, a, b are relatively prime.

1.6 Corollary.

1. If $a|c$, $b|c$ and gcd(a,b)=1, then $ab|c$.
Proof.
$a|c \Rightarrow c = au$, for some u
$b|c \Rightarrow c = bv$, for some v
As, gcd(a,b)=1, so $\exists\ x, y$ such that
$ax + by = 1$
$\Rightarrow cax + cby = c$
$\Rightarrow bvax + auby = c$
$\Rightarrow ab(vx + uy) = c$
i.e. $ab|c$.

2. If $a|bc$ and gcd(a,c)=1, then $a|b$.

Proof. $a|bc \Rightarrow bc = au$, for some u.
Also, gcd(a,c)=1
$\Rightarrow ax + cy = 1$, for some x, y.
$\Rightarrow abx + bcy = b$
$\Rightarrow abx + auy = b$
$\Rightarrow a(bx + uy) = b$
i.e. $a|b$

Exercise. Prove the following statements:
(i) gcd(a,b)= $d \Rightarrow \gcd(\frac{a}{d}, \frac{b}{d}) = 1$
Solution.
$\gcd(a, b) = d$
$\Rightarrow d = au + bv$ for some u and v
$\Rightarrow 1 = (\frac{a}{d})u + (\frac{b}{d})v$

Therefore, $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$.

(ii) $k > 0$, $\gcd(ka, kb)= k\gcd(a, b)$
Solution.

$$\begin{aligned}\gcd(ka, kb) &= (ka)u + (kb)v, \text{ for some } u, v\\ &= k(au + bv)\\ &= k \gcd(a, b)\end{aligned}$$

1.7 Theorem: If gcd(a, x)=1 and gcd(b, x)=1, then gcd(ab, x)=1.

Proof.
$\gcd(a, x)$=1 $\Rightarrow \exists\ u, v$ such that $au + xv = 1$
$\gcd(b, x)$=1 $\Rightarrow \exists\ l, m$ such that $bl + xm = 1$
Then, $(au + xv)(bl + xm) = 1$
$\Rightarrow ab(lu) + auxm + xblv + x^2vm = 1$
$\Rightarrow abK + xL = 1$, where $K = lu, L = aum + blv + vmx$
$\Rightarrow \gcd(ab, x)$=1.

1.8 Theorem: gcd(a, b)=gcd($a, b + ax$) for any integer x.

Proof. Let, d =gcd(a,b) and d' =$\gcd(a, b + ax)$
To show: $d = d'$
d =gcd(a,b) $\Rightarrow d|a$ and $d|b \ \Rightarrow d|b + ax$ and $d|a$
Also, $d'|b + ax$ and $d'|a$
Therefore, $d \leq d'$ as d' =$\gcd(a, b + ax)$

Again, $d'|a$ and $d'|1(b + ax) + (-x)a$ i.e. $d'|b$
Therefore, $d' \leq d$
Hence, $d = d'$.